

The Commutator Critic: A Grey-Box Counterfactual Update Critic for Modular Continual Learning, with Specializations to Transformers and CAROM

RelaLeap V4 proposal — companion to the V1–V3 negative-results briefing

July 2026 — v1.1, incorporating toy-scale sandbox validation

Abstract

The co-learned global causal critic program (RelaLeap V1–V3) attempted to *learn*, from decision-time summary features, a randomized paired counterfactual estimate of which module-level gradient intervention would improve a retention–plasticity tradeoff. V2 failed every discrimination and calibration gate; V3’s positive controls lost their declared orderings along the optimizer trajectory before confirmation could open. This document argues, on the basis of the General Causal-Continual-Learning (CCL) theorem, that both failures were predictable consequences of the problem formulation rather than of the causal question: (i) the summary features used were not sufficient statistics for the estimand, which to second order in the step size is an explicit functional of local gradients, curvature, and the trajectory’s linear-response propagator; and (ii) the estimand is intrinsically state-dependent, so any trajectory-invariant “declared ordering” must eventually fail, and the V3 audit criterion — not the fixture — was the defective component. We then propose the repair: a *grey-box commutator critic* whose backbone is *measured* at decision time (per-module gradient inner products, module-restricted Hessian-vector products, and a tangent-propagated linear-response term computed by forward-mode automatic differentiation along the common-random-number baseline roll-out), with a small learned head restricted to the residual that the second-order theory does not capture. Pair synergy becomes an exact cross-curvature identity, resolving the global-versus-additive comparator ambiguity; the positive control becomes a block-quadratic fixture whose ground truth is *recomputed analytically at every state*, so orderings cannot “break,” only the expansion’s validity radius can shrink — and that radius becomes a measured gate. We give the mathematics, the conceptual reframing, a software design that reuses the V3 paired-restoration machinery, specializations to Transformer head/channel modules (with HDC-footprint confusion-graph sparsification of the pair-action space) and to CAROM in its scheduled, fixed-point, and itinerant semantics (tangent propagation, implicit-function-theorem sensitivities, and measured discrete commutators respectively), and an appendix specifying a staged, gated, agent-executable experimental programme with concrete numerics chosen to preclude the shortsighted choices — unmeasured noise floors, under-powered effect sizes, leaky standardization, single-precision fixtures — that would otherwise corrupt the results. A toy-scale sandbox implementation (Section 10; a NumPy block-quadratic suite with exact closed-form truth, and a JAX MLP-plus-Adam learner with CRN paired rollouts) validates every backbone mechanism where truth is exact — including the synergy identity, the provably-aligned null, the constructive insufficient-statistics demonstration, and the margin-decay reinterpretation of the V3 audit failure — and yields concrete protocol amendments, the sharpest being that differentiating through the Adam moment recursions (exact- D) is load-bearing rather than a refinement.

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Citation caution. The handful of external citations in this note were recalled from model memory during drafting and must be verified against the literature before being relied upon or propagated.

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1 Introduction and diagnosis

1.1 What V1–V3 tried

The V-series program targets a genuinely causal object. At learner state S_t , candidate action a (attenuate one module’s incoming-task gradient, or a pair’s, before the optimizer step), ordinary action a_0 , and horizon h , the estimand is the common-random-number (CRN) paired counterfactual effect

$$\tau_h(S_t, a) = \mathbb{E}[Y_{t+h}(a) - Y_{t+h}(a_0) \mid S_t], \quad (1)$$

with Y a vector of outcomes (retained-task loss L_A , incoming-task loss L_B , and diagnostics), measured by *executing* the learner from an exact snapshot under both arms with identical randomness. This estimand, the randomized behavior policy, the exhaustive action tables, and the fail-closed gate discipline are all correct and are retained unchanged in what follows.

What failed was the estimator. V2 asked a three-branch MLP to map summary features — gradient norms, cosines, support ratios, activation statistics, Adam second moments — to τ_h , and obtained Spearman correlations below 0.065 in every family. V3 enriched the features and scaled the data tenfold, but stopped earlier: at the final audit checkpoint, every positive-control family had lost the intervention ordering *declared* for it at construction time.

1.2 Two failure theorems, informally

Both outcomes are consequences of structure the CCL theorem makes explicit.

Failure 1: the features are not sufficient statistics. To second order in the step size, τ_h is an explicit functional of the local gradients g_A, g_B , the local Hessians H_A, H_B (through their module-restricted blocks), the optimizer preconditioner, and the trajectory’s linear-response propagator (Section 3). Norms, cosines, and moment summaries quotient out precisely the curvature-weighted inner products on which the estimand depends: two states can agree on every V2 feature and have τ_h of opposite sign (Proposition 1). A critic trained on a non-sufficient statistic converges, at best, to the conditional mean over the induced equivalence class; near-zero rank correlation and amount-heads inventing their own output scale are exactly the predicted symptoms.

Failure 2: the estimand has no trajectory-invariant ordering. τ_h depends on the state through $g_A(\theta_t)$ and $g_B(\theta_t)$. As incoming-task optimization converges, the perturbation injected by a gate shrinks proportionally to $\|g_B\|$, and the retained-side alignment term can change sign as the trajectory crosses level sets of L_A (Proposition 2 gives the closed form in the quadratic case, including the geometric decay rate of the action margin). A “declared family ordering” is therefore an implicit claim of trajectory invariance that the estimand does not possess. The V3 audit did its job — it detected that the criterion, not the physics, was wrong.

1.3 The repair in one sentence

Do not train a critic to *rediscover* τ_h from summary features; *measure* the first- and second-order terms that the CCL theorem says constitute τ_h in the expansion-valid regime, and train a small model only on the residual — with a positive control whose ground truth is recomputed analytically per state, so that the only thing that can fail is the expansion’s validity radius, which is itself turned into a measured, reportable quantity.

2 Setting and estimand

Parameters $\theta \in \mathbb{R}^d$ are partitioned into M modules by disjoint index sets $I_1, \dots, I_M$; $P_m \in \mathbb{R}^{d \times d}$ is the coordinate projector onto I_m (implemented as an index mask, never a dense matrix). The learner performs preconditioned stochastic updates on the incoming task B :

$$\theta_{t+1} = T_a(\theta_t, s_t, \xi_t) = \theta_t - \eta D(s_t) G_a g(\theta_t, \xi_t), \quad g(\theta, \xi) = \widehat{\nabla} L_B(\theta; \xi), \quad (2)$$

where s_t is optimizer state, $D(s_t)$ is the (diagonal, positive) preconditioner ($D = I$ for SGD; $D_t = \text{diag}(1/(\sqrt{\widehat{v}_t} + \epsilon))$ with momentum folded in for Adam), ξ_t is minibatch randomness, and G_a is the action gate applied after backward and before the optimizer step, exactly as in V2/V3:

$$G_{a_0} = I, \quad G_{(m, \alpha)} = I - (1 - \alpha)P_m, \quad G_{(\{m, n\}, \alpha)} = I - (1 - \alpha)(P_m + P_n), \quad (3)$$

with retention factor $\alpha \in [0, 1]$ ($\alpha = 1$ is ordinary; $\alpha = 0$ freezes the module for that step). The intervention is applied at a single step t ; both arms then evolve under ordinary updates with shared $\xi_{t+1}, \dots, \xi_{t+h-1}$ (CRN). Outcomes are evaluated at $t + h$:

$$\tau_h^A(S_t, a) = \mathbb{E}[L_A(\theta_{t+h}^a) - L_A(\theta_{t+h}^{a_0})], \quad \tau_h^B(S_t, a) = \mathbb{E}[L_B(\theta_{t+h}^a) - L_B(\theta_{t+h}^{a_0})]. \quad (4)$$

We deliberately treat the outcome as a *vector* (τ_h^A, τ_h^B) and defer scalarization (the V-series utility U_h) to a separately specified decision rule; this answers Question 3 of the V3 briefing in the affirmative, and Section 3 shows the theory predicts each component separately anyway.

Throughout, write $g_A = \nabla L_A(\theta)$, $g_B = \nabla L_B(\theta)$, $H_A = \nabla^2 L_A(\theta)$, $H_B = \nabla^2 L_B(\theta)$, evaluated where stated.

3 The analytic backbone

3.1 One-step expansion

Define the *injected perturbation* of action $a = (m, \alpha)$ at step t :

$$u_t(a) := \theta_{t+1}^a - \theta_{t+1}^{a_0} = \eta(1 - \alpha) D_t P_m g_B(\theta_t), \quad (5)$$

i.e. the gated arm subtracts *less* of the incoming gradient, so it sits at the ordinary arm’s parameters plus $u_t(a)$. Writing $\delta_0 = -\eta D_t g_B$ for the ordinary step and Taylor-expanding both arms around θ_t :

Lemma 1 (One-step intervention effect). *For $L_A \in C^3$,*

$$\tau_1^A(S_t, a) = \underbrace{\langle g_A, u_t(a) \rangle}_{\text{first order: alignment}} + \underbrace{\delta_0^\top H_A u_t(a)}_{\text{drift correction}} + \underbrace{\frac{1}{2} u_t(a)^\top H_A u_t(a)}_{\text{self-curvature}} + O(\eta^3), \quad (6)$$

with all quantities evaluated at θ_t , and identically for τ_1^B with (g_B, H_B) in place of (g_A, H_A) .

Three immediate readings. First, the leading term is the preconditioner-weighted, module-restricted gradient alignment $\eta(1 - \alpha) \langle g_A, D_t P_m g_B \rangle$: protecting module m helps retention at first order exactly when the two tasks conflict there ($\langle g_A, D_t P_m g_B \rangle < 0$), which recovers gradient-surgery intuitions (PCGrad and kin; recalled from memory, verify) as the $h = 1$, first-order shadow of the causal estimand. Second, the incoming-task cost is analytic too: $\tau_1^B \approx \eta(1 - \alpha) g_B^\top D_t P_m g_B \geq 0$ for positive D_t — gating always costs incoming progress at first order, quantifying the “harm tolerance” side of the V-series policy without any learning. Third, the Adam second moments that V2 fed to the MLP as generic features enter here in their *correct structural position*: inside D_t , weighting the inner product coordinate-wise. This is the difference between giving a model the ingredients and giving it the recipe.

3.2 Horizon h : the tangent propagator

For $h > 1$ the perturbation $u_t(a)$ is transported by the linearization of the update map along the baseline trajectory. Let

$$J_k := \left. \frac{\partial T_{a_0}}{\partial \theta} \right|_{\theta_k^{a_0}} = I - \eta \left. \frac{\partial}{\partial \theta} [D_k g(\theta, \xi_k)] \right|_{\theta_k^{a_0}} \approx I - \eta D_k H_B(\theta_k^{a_0}), \quad (7)$$

where the approximation freezes the preconditioner within the differentiation (exact treatment differentiates through the moment recursions as well; both modes are implemented and compared in Stage C2). Then to first order in $\|u\|$,

$$\Delta \theta_{t+h} \approx \Phi_{t+h \leftarrow t+1} u_t(a), \quad \Phi_{t+h \leftarrow t+1} := J_{t+h-1} \cdots J_{t+1}, \quad (8)$$

and the horizon- h backbone prediction is

$$\boxed{\hat{\tau}_h^A(S_t, a) = \langle g_A(\theta_{t+h}^{a_0}), \Phi_{t+h \leftarrow t+1} u_t(a) \rangle} \quad (9)$$

and identically for $\hat{\tau}_h^B$.

Why CRN makes this the right object. Because the two arms share minibatches, optimizer state, and RNG streams, the realized paired difference *is* the response of the (stochastic) baseline trajectory to the perturbation $u_t(a)$ under the *same* noise realization. The linear response along the realized baseline path is therefore the exact first-order value of the paired estimand, with no additional Monte Carlo variance beyond that of the baseline rollout itself. The CRN design of V1–V3, built to control variance, turns out to be precisely the design under which the backbone is computable.

Computation. Equation (9) is one forward-mode tangent pass: initialize tangent $v \leftarrow u_t(a)$, and at each baseline step propagate $v \leftarrow J_k v$, which costs one Hessian-vector product (HVP) of the minibatch loss (a `jvp` through the gradient computation) plus the elementwise preconditioner action. For A candidate actions, stack tangents into a $d \times A$ matrix and batch the JVPs (`vmap` over tangent columns). Total cost: $O(A \cdot h)$ HVPs per decision state, i.e. roughly $2Ah$ extra backward-equivalents — for the Phase-0 scales below, tens of milliseconds per state on CPU, and easily amortized on GPU. Crucially, *no learned component participates*: the backbone is a measurement.

3.3 Local commutator form and the CCL connection

When retained-task replay is interleaved with incoming updates (rather than L_A merely being evaluated), the deviation between action arms over a window is governed, per the CCL theorem’s continuous specialization, by the gradient field commutator

$$[X_A, X_B](\theta) = H_B(\theta) g_A(\theta) - H_A(\theta) g_B(\theta), \quad (10)$$

restricted through the module blocks that the gate touches. The propagator form (9) is the trajectory-exact refinement of the Baker–Campbell–Hausdorff second-order bound in CCL Theorem 1: where the CCL theorem bounds forgetting by $K_{ij} h_i h_j \varepsilon_{\text{comm}}$ uniformly, the backbone *evaluates* the corresponding bilinear form on the realized trajectory. Two corollaries worth recording:

Corollary 1 (The critic estimates the confusion graph). *The matrix of backbone terms $\{\langle g_A, DP_m g_B \rangle\}_m$ and cross-curvature terms (Lemma 2) is a per-state, per-module empirical estimate of the CCL confusion structure $\delta_{c,c'}$ restricted to the module partition. A working commutator critic is therefore simultaneously the measurement instrument that the approximate categorical CCL theorem (and the CGCCT commutator loss) requires.*

Corollary 2 (Policy as commutator suppression). *The conservative policy “gate module m when the calibrated benefit is high” is, in backbone terms, “suppress the update components that load on large realized commutator terms” — i.e. the critic-driven router is an online enforcement mechanism for the CCL hypotheses, closing the loop between the theorem (modularity implies small forgetting) and the mechanism that maintains modularity.*

3.4 Pair actions: additivity at first order, synergy as cross-curvature

For the pair action, $u_t(\{m, n\}) = u_t(m) + u_t(n)$ is *linear* in the single-module perturbations, so the first-order term is exactly additive. The V2 observation that a global architecture beat the additive comparator on pair *amounts* but not pair *signs* is thereby explained: at first order there is nothing non-additive to learn. Genuine synergy is a curvature phenomenon, and it has a closed form:

Lemma 2 (Exact one-step synergy identity). *With $u_m = u_t(m)$, $u_n = u_t(n)$ as in (5),*

$$\tau_1^A(\{m, n\}) - \tau_1^A(m) - \tau_1^A(n) = u_m^\top H_A u_n + O(\eta^3) = \eta^2(1-\alpha)^2 g_B^\top P_m D H_A D P_n g_B + O(\eta^3). \quad (11)$$

Proof. All first-order and drift terms in (6) are linear in u and cancel in the difference; the quadratic term contributes $\frac{1}{2}[(u_m + u_n)^\top H_A (u_m + u_n) - u_m^\top H_A u_m - u_n^\top H_A u_n] = u_m^\top H_A u_n$. $\square$

This resolves the V3 briefing’s Question 5. The “additive independent comparator” is exactly the backbone with cross-curvature terms deleted; the “global” model is the backbone with them included; and the synergy test becomes: *does the measured cross-block $u_m^\top H_A u_n$, computable with one HVP and one dot product, predict the pair-minus-singles rollout difference?* That is a well-posed, identifiable question with a known answer in the quadratic fixture — and it dictates fixture design: the planted cross-curvature must be large enough that (11) exceeds the measured rollout noise floor by a safe factor (Appendix, Stage C1 numerics). Over horizons $h > 1$ the singles also interact through the propagator’s own curvature; the one-step identity remains the leading term, and the residual head (Section 5) owns the rest.

3.5 Preconditioner differentiation and surrogates

Two backbone modes are implemented and compared:

- **Frozen- D** : treat D_k as a constant within each step’s Jacobian (7). Cheap; exact for SGD; an approximation for Adam whose quality is an empirical question settled at probe states.
- **Exact- D** : differentiate through the moment recursions $(\hat{m}, \hat{v})$ with functional-programming AD (`torch.func`); $\approx 2\times$ the frozen cost. Used as ground truth for validating frozen- D at probe states, and as the default wherever the discrepancy gate fails.

Where full HVPs are too expensive (large Transformers), Gauss–Newton / empirical-Fisher block surrogates replace H_A in (6) and (11), and K-FAC-style Kronecker factors (recalled from memory, verify) supply per-head cross-blocks; each surrogate is admitted only after passing a cosine/magnitude agreement gate against true HVPs at probe states (Appendix, Stage C4).

4 Why the black-box critic could not work

Proposition 1 (Insufficient statistics). *Let $\varphi(S)$ be any feature map built from per-module norms of g_A, g_B , their cosines, parameter norms, activation moments, and coordinate-wise optimizer-moment summaries that are invariant under a simultaneous permutation π of coordinates within a module applied to both gradients and the corresponding moment entries. In the block-quadratic fixture of Section 6 with non-permutation-invariant within-module spectra, there exist state pairs S, S' with $\varphi(S) = \varphi(S')$ and $\tau_h^A(S, a) \tau_h^A(S', a) < 0$ for single-module actions a and horizons $h \geq 1$. Consequently any measurable critic $f(\varphi(S), a)$ satisfies*

$$\text{Var}(\mathbb{E}[\tau_h \mid \varphi]) \leq R_\varphi^2 \text{Var}(\tau_h) \quad \text{with } R_\varphi^2 < 1 \text{ bounded away from } 1,$$

and its achievable rank correlation with τ_h is bounded accordingly.

Proof sketch. Within module m , τ_h^A depends on curvature-weighted inner products of the form $\sum_{i \in I_m} w_i g_{A,i} g_{B,i}$ with weights w_i determined by the spectra of Q_A, Q_B and the propagator factors $(1 - \eta b_i)^{h-1}$; a within-module permutation of the gradient coordinates preserves norms, cosines, and moment summaries but reweights the sum, and with non-degenerate spectra one can choose sign patterns making the reweighted sum change sign. Both permuted configurations arise as genuine states of the fixture. Averaging τ over the equivalence class attenuates $\mathbb{E}[\tau \mid \varphi]$ toward zero on these classes. $\square$

Stage C1 of the appendix instantiates this constructively (a “controlled autopsy”): the V2-style MLP on V2-style features is retrained on fixture data where the backbone attains Spearman ≥ 0.95 , and its ceiling is reported. The point is not to relitigate V2 but to convert the diagnosis into a regression test: any future feature set must break the permutation invariance or it is provably capped.

Proposition 2 (No trajectory-invariant ordering; margin decay). *In the quadratic fixture with $L_B = \frac{1}{2}(\theta - b^*)^\top Q_B(\theta - b^*)$ trained by SGD, the injected perturbation obeys $\|u_t(a)\| \leq \eta(1 - \alpha) \|Q_B\| \|\theta_t - b^*\|$, and $\|\theta_t - b^*\|$ contracts geometrically at rate $(1 - \eta \lambda_{\min}(Q_B))$. Hence every action’s effect, and every between-action margin, decays geometrically along the trajectory; and the first-order retained-side term $\langle g_A(\theta_{t+h}), (I - \eta Q_B)^{h-1} u_t(a) \rangle$ changes sign whenever the trajectory crosses the corresponding level structure of L_A . No family-level ordering declared at construction time can be trajectory-invariant unless the construction pins the trajectory itself.*

The V3 audit failure is this proposition observed in the wild. The corrective is not a cleverer fixture whose ordering survives — the physics forbids it — but a fixture whose *truth is recomputed at every audited state* (Section 6), plus an explicit *no-signal regime*: when the measured between-action margin at a state falls below the rollout noise floor, the state is labeled signal-free and excluded from ordering claims rather than counted as a violation. This also answers the briefing’s Question 2: re-anchoring, if used at all, should be triggered by the measured counterfactual action margin — which the backbone supplies for free — never by retained loss or a fixed schedule.

5 The grey-box critic

5.1 Architecture

The critic is a sum:

$$\widehat{Y}_h(S_t, a) = \underbrace{\widehat{\tau}_h(S_t, a)}_{\text{measured backbone (9), incl. synergy (11)}} + \underbrace{r_\psi(\phi_h(S_t, a))}_{\text{learned residual head}}, \quad (12)$$

predicting the outcome vector (τ_h^A, τ_h^B) jointly. The residual head is deliberately small (a 2×64 MLP per outcome, five-member ensemble as in V2/V3) and is trained on $Y_h^{\text{rollout}} - \widehat{\tau}_h$, i.e. only on what the second-order theory misses: finite- $(1 - \alpha)$ effects beyond linear response, cross-arm trajectory divergence at long horizons, preconditioner nonstationarity under frozen- D , and minibatch-curvature fluctuation. Its feature vector ϕ_h contains the backbone terms themselves (first-order, drift, self- and cross-curvature, per horizon), plus phase descriptors: $\|g_A\|, \|g_B\|$, the progress ratio $L_B(\theta_t)/L_B(\theta_0)$, a 10-iteration power-method estimate of $\lambda_{\max}(DH_B)$, and the measured validity statistic of Section 5.2. Features remain decision-time observable; the anti-leakage rules of V2/V3 apply unchanged.

5.2 Validity gate: the expansion’s trust region

At a random subset of decision states (the *probe states*), both the backbone prediction and the true CRN paired rollout are computed for the full action table, and the agreement

$$\rho_h = \text{Spearman}(\widehat{\tau}_h, \tau_h^{\text{rollout}}), \quad e_h = \text{median} \frac{|\widehat{\tau}_h - \tau_h^{\text{rollout}}|}{\max(|\tau_h^{\text{rollout}}|, 3\sigma_{\text{noise}}(h))}$$

is logged as a function of horizon, attenuation strength, and phase. The *validity radius* is the largest $(h, 1 - \alpha)$ region in which $\rho_h \geq 0.8$ and $e_h \leq 0.25$. Inside the radius the backbone alone is the critic and the residual must not hurt; outside, the residual must earn its keep (gates in Section 5.4). The validity radius replaces the V3 notion of a trajectory audit against declared orderings: it is the honest version of the same instinct, measuring where the theory holds rather than insisting that a label does.

5.3 Decision rule

The conservative V-series policy shape is retained but re-expressed on vector outcomes: choose a non-ordinary action only when (i) the state’s action margin exceeds $3\sigma_{\text{noise}}$; (ii) the calibrated probability that $\tau_h^A < 0$ exceeds the frozen threshold with 4/5 ensemble agreement; (iii) the predicted incoming harm τ_h^B is below the frozen tolerance (0.02 nat retained from V3); and (iv) the state lies inside the validity radius *or* the residual head has passed its out-of-radius skill gate. Scalarization by U_h is available for reporting continuity with V1–V3 but is not the training target.

5.4 Revised gate suite

The estimation gates become:

1. **Backbone-validity gate.** $\rho_h \geq 0.8$, $e_h \leq 0.25$ inside the declared operating region, per family, on untouched confirmation seeds; states with margin $< 3\sigma_{\text{noise}}$ excluded from ordering metrics and their fraction reported.
2. **Residual-skill gate.** Inside the validity radius, adding the residual changes MSE by at most +5% (do-no-harm); outside it, the residual improves MSE over backbone-alone by $\geq 20\%$ and over the constant and support-only baselines as before. This is strictly harder and strictly fairer than V2’s “beat a constant”: the null model is now the theory.
3. **Synergy gate.** The measured cross-curvature (11) predicts pair-minus-singles rollout differences with relative error $\leq 10\%$ in the fixture and Spearman ≥ 0.6 on the real learner; the additive comparator is defined as cross-terms-deleted, ending the comparator ambiguity.
4. **Null gate.** In the provably-aligned null family (Section 6), the first-order term is nonnegative *analytically at every state*; any negative measurement exceeding $3\sigma_{\text{noise}}$ is a fixture or code defect and halts the stage. False-beneficial rate at the policy threshold $\leq 5\%$ retained.
5. **Calibration and ESS gates** retained from V3 (ten-bin ECE ≤ 0.15 on sign probabilities; audit ESS ≥ 50 per action class, ≥ 20 per module/action marginal).

6 The trajectory-stable positive control: a block-quadratic fixture

The fixture answers the briefing’s Question 1. Losses are

$$L_A(\theta) = \frac{1}{2}(\theta - a^*)^\top Q_A(\theta - a^*), \quad L_B(\theta) = \frac{1}{2}(\theta - b^*)^\top Q_B(\theta - b^*), \quad (13)$$

with stochastic gradients $g = \nabla L_B + \sigma \xi$, $\xi \sim \mathcal{N}(0, I)$, and everything run in `float64`. Because the dynamics are affine, the paired counterfactual τ_h at *any* state, action, and horizon has a closed form (a finite product of affine maps), so ground truth is *recomputed per state* — nothing is “declared,” hence nothing can be violated by the trajectory. Under Adam the dynamics are no longer affine, but exact paired rollouts remain cheap and the SGD closed form remains the analytic anchor.

The four families are realized structurally:

- **Null / harmful (provably aligned).** $a^* = b^*$, Q_A, Q_B diagonal (hence module-aligned and simultaneously diagonalizable). Then $\langle g_A, DP_m g_B \rangle = \sum_{i \in I_m} d_i q_i^A q_i^B (\theta_i - \theta_i^*)^2 \geq 0$ at *every* state for every module: gating can only cost. The V2 pathology (a null family quietly emitting 1,313 positive labels) becomes analytically impossible rather than empirically embarrassing.
- **Local-specific.** a^* and b^* differ only on I_m , with elevated Q_A diagonal there: conflict is concentrated in one module, and the per-state truth says when (and until when) protecting it helps.
- **Downstream chain.** A symmetric off-diagonal block of Q_B coupling I_m to I_n : perturbing m propagates to n through the propagator $(I - \eta DQ_B)^{h-1}$, so coordinated protection is favored at moderate h — verifiable from the closed form.
- **Synergistic pair.** A symmetric off-diagonal block C of Q_A between I_m and I_n , sized so the one-step synergy (11) satisfies $|u_m^\top Q_A u_n| \geq 5 \sigma_{\text{noise}}(1)$ at typical states (numerics in Stage C1).

The fixture also hosts the two propositions of Section 4 as executable demonstrations: the insufficient-statistics construction (permuted-state pairs) and the margin decay curve.

7 Specialization to Transformers

7.1 Module partition and cost

Following CCL §7 and the CGCCT module inventory, modules are attention heads and feed-forward channel groups; for GPT-2 small that is 144 heads plus FF blocks, and for the Phase-0 model of Stage C4 it is 8 heads plus 2 FF groups. Per-module gradients are obtained from the same backward pass via parameter grouping (no extra cost); the first-order backbone term for all M single-module actions is then M masked inner products — essentially free. The propagator costs one batched JVP pass per decision state (Section 3.2); cross-curvature terms cost one HVP per participating module. For large models, empirical-Fisher / GGN block surrogates and K-FAC factors stand in for HVPs, gated by probe-state agreement.

7.2 Sparsifying the pair-action space with HDC footprints

Pair actions scale quadratically in M ; at 144 heads that is untenable and also unnecessary. The HDC-CGCCT synthesis supplies the fix: encode each module’s do-footprint as a hypervector (Eq. 5 of that note), build the module-level confusion graph by approximate-nearest-neighbor search over footprint codes, and instantiate pair actions (and cross-curvature measurements) *only* for edges of the confusion graph. The backbone then feeds measured commutator terms back as edge weights, so the graph and the critic co-refine — the “sparse confusion graph” of the approximate categorical CCL theorem, estimated online.

7.3 Retained-gradient estimation and CMCP

On real learners g_A is estimated from replay. Duplicate or derived replay packets inflate the effective precision of that estimate in exactly the way the CMCP calibration study characterized; the typed innovation/provenance guard (weight-zeroing byte-identical and declared-descendant packets) is therefore applied to the replay stream feeding the g_A estimator, with the total-mass normalization recommended in that report. This is the one place the CMCP machinery enters the critic, and it enters as plumbing, not as a hypothesis under test.

8 Specialization to CAROM

CAROM (Content-Addressable Recurrent Operator Machine) separates control from data: opaque command tokens route computation over a small library of reusable learned operators that read, transform, gate, and write a slot workspace over recurrent steps, with scheduled, fixed-point (equilibrium), and itinerant execution semantics, and a PC/EqProp local-learning variant (PC-CAROM). This separation makes CAROM, if anything, a *better* host for the commutator critic than a Transformer, because the module partition is architectural rather than discovered: operator weight blocks $\{W_k\}$, the control system, and per-slot gating parameters are natural modules, and a context’s support is literally the set of operators its programs invoke.

- **Scheduled / itinerant (BPTT-trained) semantics.** The update map is an unrolled program; the tangent propagator of Section 3.2 applies verbatim, with J_k the Jacobian of one recurrent step. Actions attenuate a specific operator’s gradient during adaptation to a new context; the backbone predicts retained-context effects per operator. Because programs are short (length ~ 6 in the sandbox series), the propagator is cheap and well-conditioned.
- **Fixed-point semantics.** At an equilibrium z^* with $F(z^*, \theta) = 0$, the implicit function theorem gives the sensitivity $\partial z^* / \partial \theta = -(\partial F / \partial z)^{-1} \partial F / \partial \theta$; the backbone’s linear-response term is obtained by solving one linear system per action (conjugate gradients / Neumann truncation, as in deep-equilibrium practice; recalled from memory, verify), validated against nudged finite differences. These equilibrium intervention Jacobians are, not coincidentally, the do-influence estimates that causal coding wants for its gates — the critic and the causal-coding machinery share their most expensive measurement.
- **Non-gradient components.** Where dynamics are genuinely non-gradient — Lotka–Volterra mode arbitration, discrete routing — the smooth backbone does not apply, and the *discrete* CCL commutator does: measure $d(U_i U_j(\Theta), U_j U_i(\Theta))$ directly by paired CRN rollouts from snapshots, with d the max over per-context evaluation losses. The V3 runner is, unmodified,

a discrete-commutator estimator; Stage C5 tests whether measured commutators predict realized forgetting across context pairs, which is the CAROM-native form of the critic’s claim.

A natural downstream hook, out of scope here but worth one sentence: the measured module-level commutator/confusion graph exports directly to Atomspace as typed links, making the critic’s state legible to MeTTa/PLN-level meta-reasoning about what to protect — the same interface pattern as the causal-fibres symbolic head.

9 Software design

9.1 Placement and layout

New code lives in the Relaleap repository as a sibling of the V1–V3 worktree, reusing its paired snapshot/restore, propensity logging, seed discipline, and RUN-record conventions unchanged:

Path	Contents
src/relaleap/critic/partition.py	ModulePartition (index sets; masked ops; <i>no dense projection</i>)
src/relaleap/critic/backbone.py	one-step expansion; tangent propagator (frozen- D / exact-cross-curvature via HVP; GGN/K-FAC surrogates)
src/relaleap/critic/fixtures/quadratic.py	family constructors; closed-form τ_h ; margin and noise-floor
src/relaleap/critic/residual.py	feature builder; per-(family, horizon) train-only standardization
src/relaleap/critic/gates.py	validity, residual-skill, synergy, null, calibration, ESS gates
src/relaleap/critic/confusion.py	footprint codes; ANN confusion graph; pair-action instantiation
scripts/run_commutator_critic_c{0..6}.py	stage runners (Appendix)
tests/test_critic*.py	the test inventory below

9.2 Interfaces

`Backbone.predict(state, actions, horizons) -> {tauA_hat, tauB_hat, terms}` returns the prediction *and* the named terms (first-order, drift, self-curvature, cross-curvature, propagated), so that gates, residual features, and reports consume the same decomposition; **terms** is the anti-black-box contract of the whole design. `QuadraticFixture.exact_tau(state, action, horizon)` returns closed-form truth. `PairedRunner` is the V3 runner; the only addition is a callback slot that lets the backbone ride along the baseline rollout (propagating tangents) without perturbing it. All stochastic components take explicit `torch.Generators`; nothing touches global RNG state.

9.3 Numerical hygiene (binding on all stages)

1. **Precision.** The quadratic fixture and all validity-gate arithmetic run in `float64`. Transformer/CAROM stages run `float32` with `float64` accumulation for inner products and losses. No `tf32`, no AMP, anywhere in Phase 0.
2. **Step-size discipline.** Enforce $\eta \lambda_{\max}(DH) \leq 0.5$ (power-method estimate, 10 iterations, 3 probes) before any run; log the margin. Half the historical “mysterious” fixture instability is edge-of-stability dynamics that this check excludes by construction.

3. **Measured noise floors.** $\sigma_{\text{noise}}(h)$ is estimated from ≥ 64 replicate CRN pairs at ≥ 20 probe states per family before training data are generated, never assumed. All effect-size, margin, and validity thresholds are expressed in units of the measured floor.
4. **Effect-size precheck.** Before generating a stage’s training data, verify that the median $|\tau_h^A|$ over probe states and non-ordinary actions is $\geq 5 \sigma_{\text{noise}}(h)$ at the operating (h, α) ; if not, strengthen α or the planted conflict and re-run the precheck. This single rule would have prevented the V2 tiny-utility regime.
5. **No leaky standardization.** Target standardization (residual head only) is per (family, horizon), fitted on training seeds, applied frozen; sign targets are never standardized.
6. **Paired statistics only.** All comparisons between arms, methods, or ablations are paired per (state, action, seed); minimum 3 seeds, 5 preferred; report per-seed numbers.
7. **HVP verification.** Every HVP pathway is checked against central finite differences with step $10^{-3} \sqrt{d} \|v\|^{-1}$ -scaled perturbations: relative error $\leq 10^{-6}$ (`float64`) and $\leq 10^{-3}$ (`float32`) at 20 random (θ, v) pairs, as a standing `pytest` fixture, not a one-off.

9.4 Test inventory

(i) partition masks: gate algebra identities $G_{a_0} = I$, idempotence, disjointness; (ii) HVP vs finite differences as above; (iii) tangent propagator vs closed-form quadratic truth to $\leq 10^{-8}$ relative error in `float64`; (iv) frozen- D vs exact- D discrepancy report; (v) paired restoration bit-exactness (import the V3 test); (vi) determinism: two runs, same seeds, identical artifacts byte-for-byte; (vii) null-family analytic nonnegativity checked at every generated state; (viii) synergy identity (11) vs pair-minus-singles closed form; (ix) gate implementations against hand-computed miniature cases.

10 Toy-scale sandbox validation and lessons learned

Before committing agent budget to Stages C0–C6, the mechanisms were exercised end-to-end at toy scale in two self-contained scripts (`quadratic_demo.py`, NumPy, `float64`; `mlp_demo_jax.py`, JAX, `float64`; CPU, minutes). The sandbox is not a substitute for the staged programme — scales, learners, and statistics are deliberately miniature — but it settles the question a referee should ask first: *do the claimed identities and estimators actually behave as derived, in running code, where truth is exact?* They do, and the exercise produced protocol amendments recorded below and threaded into the appendix.

10.1 Setup

Quadratic suite. The fixture of Section 6 at $d = 64$, $M = 4 \times 16$, SGD $\eta = 0.1$, gradient noise $\sigma = 10^{-3}$, gates applied to the *realized stochastic* gradient, CRN paired rollouts as in V2/V3. The backbone accesses the fixture only through decision-time-legal oracles (g_A , g_B , HVPs) — never the matrices — mimicking the production API. All four families run; ground truth is the closed-form counterfactual, recomputed per state.

MLP suite. An $8 \rightarrow 32 \rightarrow 32 \rightarrow 4$ `tanh` MLP with three modules (layer 1, layer 2, head), pretrained on a teacher task A (600 Adam steps), then adapted to a conflicting task B (150

steps, fresh Adam state), with decision states every 10 updates, actions gating one module (or the layer-1/layer-2 pair) at $\alpha \in \{0.5, 0.1\}$, horizons $h \in \{1, 5, 10, 25\}$, and truth from CRN paired rollouts. Conflict families: B_1 input-permutation ($y_B = f_A(Px)$; conflict planted in layer 1), B_2 output-shift ($y_B = f_A(x) + c$; head), and a stress family (B_1 at $10\times$ learning rate). The backbone computes the *exact* injected perturbation of the full optimizer state (θ, m, v) — one gated versus one ordinary step from the snapshot — and propagates it by forward-mode `jax.jvp` through the *entire* Adam update (exact- D), alongside the frozen- D approximation of Eq. (7).

10.2 Results: quadratic suite (exact truth)

check	result	gate analogue
CRN rollout vs closed form	rel. err $\leq 2.6 \times 10^{-8}$	C0 self-test
backbone Spearman (all families, $h \leq 20$)	0.998–1.000	C1 Gate 1
backbone median rel. err	0.09%–0.56%	C1 Gate 1
decision-time (preview) backbone Spearman	0.998–1.000	—
synergy identity vs pair–singles	rel. err 0.0000; controls $\leq 10^{-15}$	C1 Gate 2
synergy effect size over noise floor	$22 \times / 16 \times / 9 \times$ at $h = 1/5/10$	precheck
null: min first-order alignment	$+1.85 \times 10^{-4} > 0$, all states	C1 Gate 3
null: false-beneficials beyond 3σ	0	C1 Gate 3
margin decay (640-step trajectory)	$0.188 \rightarrow 9 \times 10^{-5}$, geometric	Prop. 2
no-signal fraction, first→last quartile	$0.00 \rightarrow 1.00$	Prop. 2

The backbone’s entire residual error in the quadratic is the omitted second-order term $\frac{1}{2}\Delta^\top Q_A \Delta$ — visible, bounded, and exactly the object assigned to the residual head. The margin-decay row is Proposition 2 observed under a microscope: the fitted decay rate ($-0.0100/\text{step}$) sits inside the worst-case bound ($\ln(1 - \eta\lambda_{\min}) = -0.0054$), and the audit-relevant quantity — the fraction of states with no resolvable signal — goes from zero to one along the trajectory. Nothing “breaks”; the estimand drains.

The insufficient-statistics proposition ran constructively: 400 states were Gauss–Newton-projected onto identical V2-style feature vectors (mismatch $< 10^{-10}$). Exact τ_1 coincides across the class to 5×10^{-15} — the first-order-plus-uniform-curvature component *is* feature-determined, as the theory says — while exact τ_{10} spans $[-3.2 \times 10^{-4}, +9.0 \times 10^{-3}]$ with 7 sign flips against the reference state. Cross-instance (train on four wide-spectrum fixture instances, test on a fifth, $h = 50$ where propagator reweighting dominates), ridge and 5-NN on V2-style features reach Spearman 0.21 and 0.46 respectively; the backbone reaches 1.00 on the same test set, because it measures the test instance’s curvature at runtime rather than hoping the features encoded it.

10.3 Results: MLP + Adam (truth = CRN paired rollouts)

Backbone Spearman versus rollout truth, margin-filtered:

family / mode	$h = 1$	$h = 5$	$h = 10$	$h = 25$
B_1 input-perm., exact- D	1.000	1.000	1.000	1.000
B_1 input-perm., frozen- D	1.000	0.988	0.972	0.923
B_2 output-shift, exact- D	1.000	1.000	1.000	1.000
B_2 output-shift, frozen- D	1.000	0.979	0.925	0.852
B_1 stress ($10\times$ lr), exact- D	1.000	0.999	0.998	0.997
B_1 stress ($10\times$ lr), frozen- D	1.000	0.156	−0.054	0.203

Median relative error of exact- D stays in 0.03%–1.9% throughout, including the stress family. The one-step synergy identity, evaluated on the real network via one HVP, matches the measured pair-minus-singles difference at Spearman 0.996–1.000 (median rel. err 0.7%–1.7%). The analytic incoming-harm claim holds where stated: the fraction of actions with $\tau^B \geq 0$ is 0.98–1.00 at $h = 1$ (first order: gating always costs incoming progress) and decays at long horizons where second-order effects enter — the vector-outcome design earning its keep.

10.4 Lessons learned

1. **The effect-size precheck caught a live failure.** The first synergy plant — a random-direction cross-block, as originally specified — landed at $0.1\times$ the measured noise floor: a faithful reproduction of V2’s tiny-utility regime, in fresh code, on the first try. The precheck flagged it; the repair (a rank-1 cross-block *aligned with the early-phase incoming gradient*, measured on the early trajectory slice) brought the plant to $9\text{--}22\times$ the floor. The C1 numerics are amended accordingly, and the episode is the strongest available argument for keeping the precheck mandatory and unfrozen-in-spirit: it is the rule that distinguishes “the mechanism is absent” from “the fixture whispered.”
2. **Exact- D is load-bearing under Adam, not a refinement.** At learner-typical $\eta = 10^{-3}$, frozen- D degrades gracefully (0.85–0.97 at $h \geq 10$); at $\eta = 10^{-2}$ it collapses to noise while exact- D is essentially unaffected. Differentiating through the moment recursions is where most of the horizon-transport fidelity lives. Amendment: exact- D is the *default* for Adam learners in C2/C4; frozen- D is an optimization admitted only by the probe gate, with the expectation that it fails at practical learning rates.
3. **A decision-time backbone exists.** The deployment-legal variant — a noise-free *preview* rollout standing in for the realized future baseline — matched validation-mode Spearman to three decimals in the quadratic. The critic may not need the realized future at all; C2 gains a preview-mode comparison arm to test this on real learners, where minibatch curvature fluctuation could yet separate them.
4. **Measure, don’t declare — confirmed on a real learner.** In B_1 the conflict is *planted* in layer 1, yet the realized retention benefit at $h = 10$ orders head $>$ layer 2 $>$ layer 1: Adam distributes the interfering update over all modules, and the causally efficacious gate is not the architecturally intended one. Any residual temptation to reintroduce declared per-family orderings for real-learner audits should end here.
5. **The failure propositions are now regression tests.** Both propositions of Section 4 exist as constructive, seed-stable code (feature-identical sign-flip states; the margin-decay /

no-signal curve). They move from post-mortem narrative to standing `pytest` fixtures: any proposed feature set or audit criterion must pass through them.

6. **The learnable residue is small at toy scale.** With exact injection and exact- D transport, what remains for a residual head is 0.03–1.9% relative — consistent with the possibility, flagged in Section 11, that the programme’s honest landing is the backbone as a measurement instrument, with the learned component thin or empty. C3’s do-no-harm gate is therefore not a formality; it is the likely binding constraint.

11 What would falsify this proposal

The proposal is fail-closed like its predecessors, and its falsifiers are sharp. If the backbone fails its own validity gate *inside* the fixture (Stage C1), the implementation is wrong — stop. If the backbone validates on the fixture but shows no validity region at all on the real MLP learner at $h \leq 5$ and strong attenuation (Stage C2), then second-order local theory genuinely does not govern short-horizon intervention effects in trained networks at Phase-0 scale — a real and publishable negative that would also, notably, constrain the quantitative reach of the CCL theorem’s smooth specialization. If the backbone validates but the residual head cannot beat it anywhere outside the radius (Stage C3), the honest conclusion is that the *learnable* part of the critic problem is empty at this scale, and the program should ship the backbone as a measurement instrument (for CGCCT commutator losses and confusion graphs) rather than as a learned critic — which would itself be a useful landing.

A Agent experiment programme: stages C0–C6

This appendix is written for autonomous coding agents. Run stages in order; each has purpose, procedure, gates, numerics, budget, and stop conditions. **A cleanly measured null is a successful outcome; the only failure is a corrupted positive.** Fix only what blocks execution; record every deviation in one line. Frozen choices (A.8) may never be changed. All work through C3 is local CPU, matching the V-series discipline; C4 is the first GPU stage and is gated on C1–C2 passing.

Where a number below is given as a default, use it; where a range is given, sweep it and report the sweep, do not silently pick one point. The numerics were chosen for statistical power and conditioning, not convenience; if a budget forces reduction, reduce the number of *sweeps*, never the number of *seeds*, *probe states*, or *noise-floor replicates*.

A.1 Stage C0: infrastructure and self-tests (CPU, ≤ 2 h)

Purpose. Stand up the library and prove the measurement plumbing before any science. Reference implementations of the expected numerical behavior exist (`quadratic_demo.py`, `mlp_demo_jax.py`, Section 10); C0 ports their checks into `pytest` with the tolerances below, including the two constructive failure-proposition demos as standing regression tests (Lesson 5).

Procedure and gates.

1. Implement `partition.py`, `backbone.py`, `fixtures/quadratic.py`; run the full test inventory of §9.

2. HVP gate: relative error vs central finite differences $\leq 10^{-6}$ (**float64**, $d = 512$ quadratic) and $\leq 10^{-3}$ (**float32**, the V2 three-branch MLP), 20 random (θ, v) pairs each.
3. Tangent-propagator gate: vs closed-form quadratic τ_h , relative error $\leq 10^{-8}$ at $h \in \{1, 5, 20\}$, 10 states, all 8 single actions, **float64**, SGD.
4. Paired-restoration gate: import and pass the V3 exact-restoration test unchanged (learner, Adam state, RNG streams bit-identical after snapshot/restore).
5. Determinism gate: two full C0 runs with identical seeds produce byte-identical JSON artifacts.

Stop if any gate fails after two fix attempts.

A.2 Stage C1: quadratic fixture — backbone validation and controlled autopsy (CPU, $\approx 4\text{--}8$ h)

Purpose. Validate the backbone against exact truth; demonstrate the two failure propositions constructively; calibrate the synergy design rule.

Fixture numerics (defaults; freeze after this stage).

- Dimension $d = 512$; $M = 8$ modules of 64; **float64**.
- Spectra: Q_A, Q_B diagonal entries log-uniform on $[0.05, 1.0]$ (condition number 20); optimizer SGD, $\eta = 0.1$ (so $\eta\lambda_{\max} = 0.1$, comfortably inside the 0.5 ceiling); a second arm with Adam($\eta = 10^{-2}$, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$).
- Gradient noise $\sigma \in \{10^{-3}, 10^{-2}\}$ (two arms).
- Families: null ($a^* = b^*$, both diagonal — aligned by construction); local-specific ($b^* - a^*$ supported on I_3 with $\|b^* - a^*\| = 2$, Q_A diagonal on I_3 scaled $\times 3$); chain (Q_B cross block between I_2, I_5 with spectral norm 0.3); synergy (Q_A cross block between I_4, I_7 : *rank-1 and aligned with the early-phase incoming gradient direction*, spectral norm 0.4, with the precheck measured on the early trajectory slice — the sandbox showed a random-direction block lands at $0.1\times$ the noise floor (Lesson 1); verify $\geq 5\sigma_{\text{noise}}(1)$ in the effect-size precheck and, if violated, raise toward 0.5 or restrict the state slice earlier, never lower the noise).
- Trajectories: 400 SGD steps from $\theta_0 \sim \mathcal{N}(a^*, I)$; decision states every 2 steps (200 per family per seed); horizons $h \in \{1, 2, 5, 10, 20, 50\}$; attenuations $\alpha \in \{0.5, 0.1\}$; action table: ordinary + 8 moderate + 8 strong + the planted pair and 4 random control pairs.
- Noise floors: 64 CRN replicate pairs at 20 probe states per family per horizon per σ arm (standard error of the floor estimate $\approx \sigma_{\text{noise}}/\sqrt{128}$, i.e. $\leq 9\%$ relative).
- Seeds: 5 (train 3 / confirmation 2, disjoint), fixture regenerated per seed.

Gates.

1. Backbone vs exact truth: relative RMSE $\leq 5\%$ and Spearman ≥ 0.95 , per family and horizon, restricted to states with margin $\geq 3\sigma_{\text{noise}}(h)$; report the excluded (no-signal) fraction and its growth along the trajectory — this curve is the empirical form of Proposition 2 and belongs in the report, not in the failure column.

2. Synergy: measured $u_m^\top Q_A u_n$ matches pair-minus-singles exact truth within 10% relative at $h = 1$ and within 25% at $h = 5$; the 4 random control pairs show synergy $\leq \sigma_{\text{noise}}$.
3. Null: analytic nonnegativity of the first-order term at every generated state (100%, exact); rollout false-benefit at the V3 policy threshold = 0%.
4. Controlled autopsy (diagnostic, no pass/fail): retrain the V2-style MLP on V2-style features against exact τ_h on this fixture (3,000 train / 1,000 val transitions, Adam 10^{-4} , 2,000 epochs, early stop patience 200); report its Spearman ceiling next to the backbone's, and exhibit ≥ 10 permuted-state pairs with identical features and opposite-sign truth (Proposition 1 made executable).

Stop the programme if Gate 1 fails: the backbone implementation or theory is wrong at the only place where truth is exact.

A.3 Stage C2: real MLP learner — backbone against paired rollouts (CPU, $\approx 6\text{--}12$ h)

Purpose. Establish the validity radius on the V2/V3 three-branch MLP test bed with real Adam state, real activations, real minibatches.

Numerics.

- Learner and families: the V3 fixture and behavior policy unchanged (50/25/15/10 action mix, propensities logged), Adam $\eta = 10^{-4}$, $\lambda_B = 1$, interval 10 — continuity with the V-series is the point.
- **Effect-size precheck (mandatory, before data generation).** At 30 probe states per family, measure $\sigma_{\text{noise}}(h)$ (64 CRN replicates) and the median $|\tau_h^A|$ over strong actions for $h \in \{1, 2, 5, 10\}$. Select the operating pair (h^*, α^*) as the smallest h with median effect $\geq 5\sigma_{\text{noise}}$ at $\alpha = 0.1$; if none qualifies, strengthen the planted conflict (scale the family's conflicting-target displacement by $1.5\times$, iterate at most 3 times) and record each iteration. Do not proceed on sub- 5σ effects under any circumstances.
- States: 300 train / 100 calibration / 600 confirmation per family (V3 scale), exhaustive action tables at train states, randomized audit slice as in V3; seeds disjoint as in V3.
- Backbone modes: *exact-D is the default* (Lesson 2: frozen-D collapsed at $10\times$ learning rate in the sandbox while exact-D held at Spearman ≥ 0.997). Frozen-D may be substituted as an optimization only if it passes acceptance at 50 probe states: cosine ≥ 0.95 and magnitude ratio in $[0.5, 2]$ against exact-D. Add a *decision-time preview* arm (noise-free preview rollout in place of the realized baseline; Lesson 3) and report its agreement alongside.
- Power-method λ_{max} estimates: 10 iterations, 3 probes, logged per state.

Gates. On untouched confirmation seeds, backbone-only: Spearman ≥ 0.5 vs rollout truth per family at (h^*, α^*) among margin-qualified states; sign AUROC ≥ 0.75 where two-class; synergy Spearman ≥ 0.6 for planted pairs; null false-benefit $\leq 5\%$; ESS gates as in V3. Report the full validity-radius map (ρ_h, e_h) over $h \in \{1, 2, 5, 10, 25\}$, $\alpha \in \{0.5, 0.1\}$. **Stop** (and write the negative up — it is a substantive result about second-order theory in trained networks) if no (h, α) cell passes with $h \leq 5$.

A.4 Stage C3: residual head (CPU, $\approx 4\text{--}8$ h)

Purpose. Test whether learning adds anything beyond the theory, on the honest null model.

Numerics. Residual MLP 2×64 , GELU, five-member ensemble, AdamW 3×10^{-4} , weight decay 10^{-2} , batch 256, ≤ 500 epochs, early stop on calibration seeds (patience 50); train on the C2 train states’ exhaustive tables ($\approx 12,000$ transitions across families), targets $Y^{\text{rollout}} - \hat{\tau}$, per-(family, horizon) train-only standardization; sign probabilities isotonic-calibrated on calibration seeds only, exactly as in V2/V3.

Gates. Inside the C2 validity radius: MSE change $\leq +5\%$ (do-no-harm). Outside it (specifically $h = 25$, and any failed C2 cells): MSE improvement $\geq 20\%$ over backbone-alone *and* over constant and support-only baselines; sign ECE ≤ 0.15 ; ensemble agreement statistics reported. If the residual passes nothing, ship the backbone as instrument (see falsification section) — that outcome is a deliverable, not a failure.

A.5 Stage C4: small Transformer (first GPU stage, ≤ 12 GPU-h)

Purpose. The smallest honest Transformer instantiation: per-head modules, surrogate curvature, confusion-sparsified pairs.

Numerics.

- Model: 2-layer causal Transformer, $d_{\text{model}} = 128$, 4 heads/layer (8 head modules), $d_{\text{ff}} = 512$ (2 FF modules; $M = 10$), vocab 64, sequence length 64, dropout 0, `float32` with `float64` reductions.
- Data: Task A = PCFG synthetic grammar (reuse the CGCCT Stage-P1 generator), 5M tokens, pretrain 8,000 AdamW steps, $\eta = 3 \times 10^{-4}$, $\beta = (0.9, 0.999)$, weight decay 0.01, batch 32, cosine decay to 3×10^{-5} , gradient clip 1.0. Task B = terminal-remapped grammar (systematic interference), 400 adaptation updates at $\eta = 10^{-4}$, clip 1.0.
- Retained evaluation: a frozen 200-batch Task-A eval set; g_A estimated as the mean over an 8-batch replay draw per decision state, with CMCP typed deduplication if a replay buffer (rather than fresh draws) is used.
- Decision states: every 5 adaptation updates (80 per seed); horizons $h \in \{1, 5, 10, 25\}$; $\alpha \in \{0.5, 0.1\}$; actions: ordinary, 10 singles, and the top-5 confusion-graph pairs (footprint codes per §7, $D = 8,192$ bipolar, ANN over the 45 candidate pairs is trivial at this scale but the machinery must run end-to-end because C5/scale-up depend on it).
- Curvature: true HVPs are affordable at this scale — use them; also compute the empirical-Fisher block surrogate at every state and report its agreement (cosine, magnitude ratio) as the calibration that a later GPT-2-small stage will rely on. Surrogate admission threshold for future use: cosine ≥ 0.8 , magnitude ratio in $[0.5, 2]$, at ≥ 30 states.
- Noise floors and effect-size precheck exactly as in C2 (30 probe states, 64 replicates); seeds 3 minimum, per-seed reporting.

Gates. Backbone Spearman ≥ 0.4 vs paired rollout truth at $h \leq 10$ among margin-qualified states, per seed; synergy Spearman ≥ 0.5 on confusion-graph pairs vs measured pair-minus-singles; frozen- D acceptance as in C2; residual head (retrained on Transformer data with C3 hyperparameters) subject to the same do-no-harm / earn-your-keep gates. Report, descriptively,

which heads carry persistent conflict terms — this table is the input the CGCCT commutator loss has been waiting for.

A.6 Stage C5: CAROM track (CPU or ≤ 4 GPU-h)

Purpose. The three CAROM semantics, in increasing order of novelty.

Numerics. Reuse the CAROM sandbox configuration: operator library $K = 8$, 16 slots, program length 6, 4 context families; BPTT-trained scheduled variant as substrate (i); fixed-point variant as substrate (ii); itinerant/LV arbitration as substrate (iii).

1. *Scheduled:* operator-level gating actions ($M = K + 1$ counting the control system as a module); tangent propagator through the unrolled program; gates as in C2 with the same margin discipline; 60 decision states $\times$ 3 seeds, horizons $\{1, 5, 10\}$.
2. *Fixed-point:* IFT sensitivities via conjugate gradients (tolerance 10^{-6} , max 200 iterations, warm-started); validation against nudged finite differences (nudge 10^{-3} , central): relative error $\leq 10\%$ at 20 probe equilibria before any effect estimates are reported.
3. *Discrete commutators:* 500 paired CRN measurements of $d(U_i U_j, U_j U_i)$ across the 6 context pairs from snapshots along training, $d = \max$ over per-context eval-loss differences. **Gate:** Spearman ≥ 0.5 between measured commutator magnitude and realized forgetting when the contexts are subsequently trained sequentially (both orders, paired). This is the CAROM-native critic claim; if it fails while (1) passes, the discrepancy localizes the non-gradient components as the obstruction — report which.

A.7 Stage C6: policy pilot (only if C2 and C4 pass)

The conservative V-series policy, re-expressed on vector outcomes and the validity radius (§5), evaluated on the C2 test bed first: retention vs ordinary and support-score controls, incoming loss within 0.02 nat, positive paired bootstrap (10^4 resamples), superiority to frequency-matched random, $\geq 50\%$ of oracle improvement (oracle from exhaustive tables), credit-wavefront and effective-rank diagnostics preserved. New protocol version, fresh seeds, thresholds frozen before the first confirmation seed is opened — V-series discipline unchanged.

A.8 Frozen choices and allowed knobs

Never change: the estimand (1) and CRN pairing; the gate definitions and thresholds once C1 completes; the validity-radius definition ($\rho \geq 0.8$, $e \leq 0.25$, margin $\geq 3\sigma$); the effect-size precheck rule ($\geq 5\sigma$, at most 3 strengthening iterations); the behavior-policy mix; `float64` for the fixture; the per-seed reporting requirement; the do-no-harm bound on the residual. **May change with a one-line note:** horizon and α grids; residual width/depth within $[1, 4] \times [32, 128]$; learning rates (halve on divergence, clip 1.0); CG tolerances; footprint dimension D ; number of confusion-graph pairs [3, 10].

A.9 Budgets and global stop conditions

C0–C3 are CPU-only; C4 ≤ 12 GPU-hours; C5 ≤ 4 GPU-hours; hard programme ceiling 3 GPU-days including reruns. Stop and report (partial results are a deliverable) if: C0 or C1 Gate 1 fails; C2 yields no valid cell at $h \leq 5$ and the operator directs a halt; cumulative compute exceeds the ceiling; or any fix would touch a frozen choice.

A.10 Report format

Produce `RESULTS.md` containing: environment and pinned versions; per-stage gate outcomes with measured numbers beside thresholds; the validity-radius maps (C2, C4) as tables over (h, α) ; the no-signal fraction curves; the synergy identity check; the controlled-autopsy Spearman ceiling next to the backbone’s; the frozen- D /exact- D and HVP/Fisher agreement tables; per-seed numbers throughout; every deviation in one line; and a closing paragraph answering, in order: did the backbone validate where truth is exact; what is the validity radius on real learners; does learning add anything beyond the theory; do measured commutators predict CAROM forgetting; and what single follow-up is most informative. Attach all JSON artifacts and the commit SHAs, following the V-series provenance conventions.

References

- Goertzel, B. *Causal Coding and the General Causal-Continual-Learning Theorem*. Manuscript, December 2025.
- RelaLeap project. *Co-learned Global Causal Critic: Idea, Tests, Negative Results, and Questions for External Review*. Technical briefing, July 26, 2026.
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- Goertzel, B. *Hypervector Frames for Category-Guided Causal Coding*. Manuscript, July 2026.
- CAROM series working notes, including *From Scheduled Execution to Itinerant Dynamics* and *PC-CAROM*, July 2026.
- Commutator-critic sandbox: `quadratic_demo.py`, `mlp_demo_jax.py`, with `RESULTS.md` and JSON artifacts. Companion code to this note, July 2026.
- Yu, T., et al. *Gradient Surgery for Multi-Task Learning (PCGrad)*. NeurIPS 2020. (Recalled from model memory; verify before use.)
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- Martens, J., Grosse, R. *Optimizing Neural Networks with Kronecker-factored Approximate Curvature (K-FAC)*. ICML 2015. (Recalled from model memory; verify before use.)